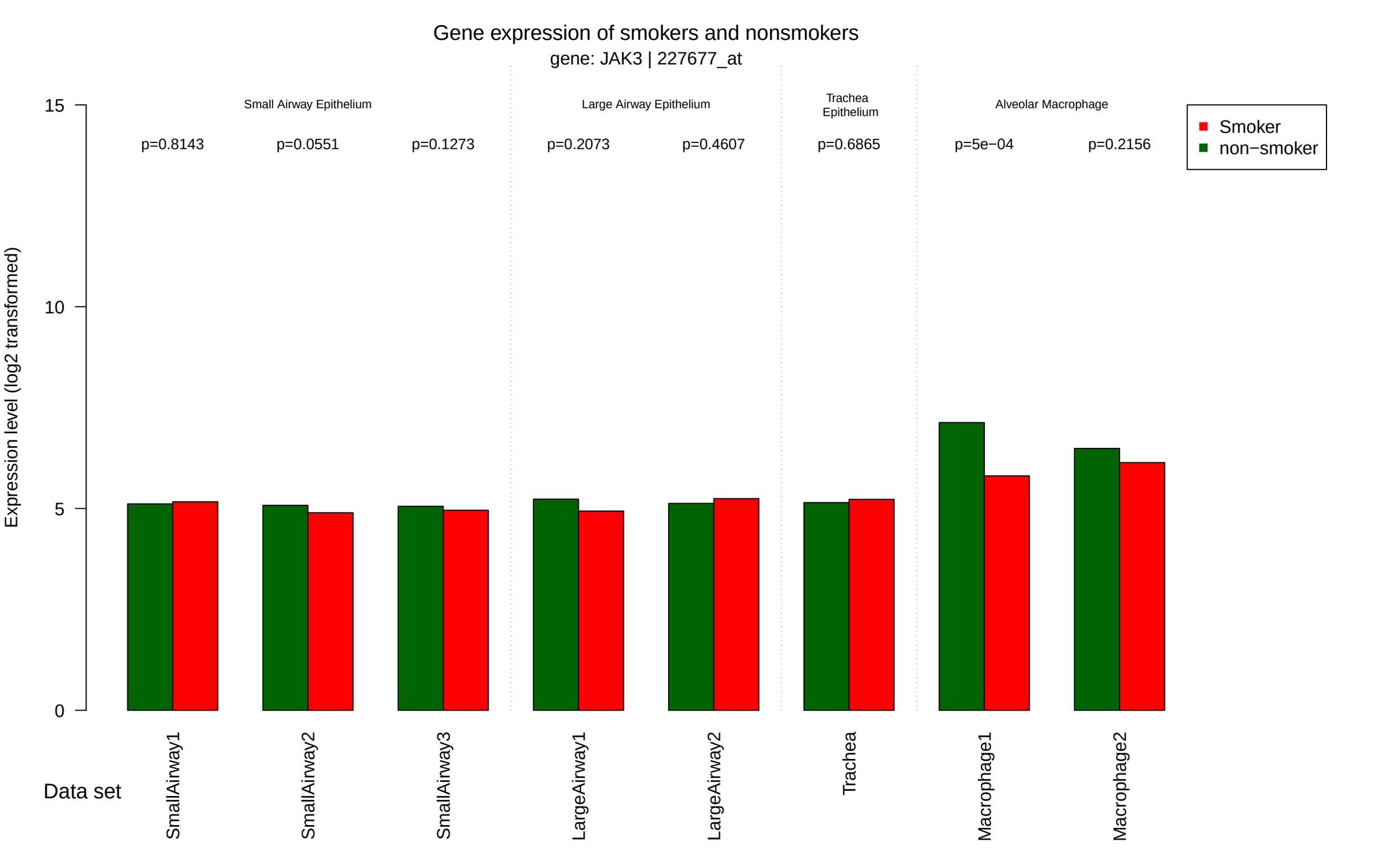
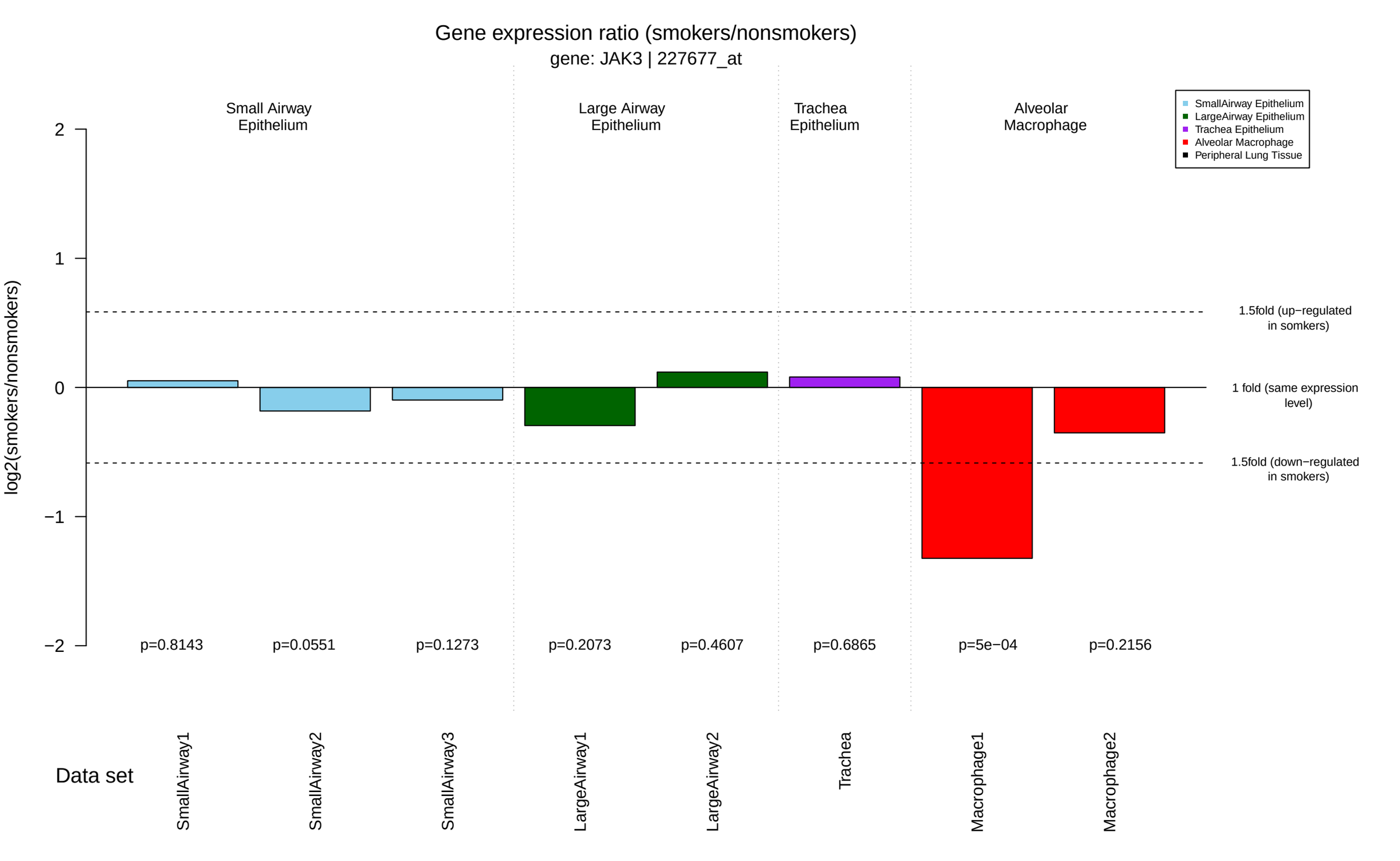
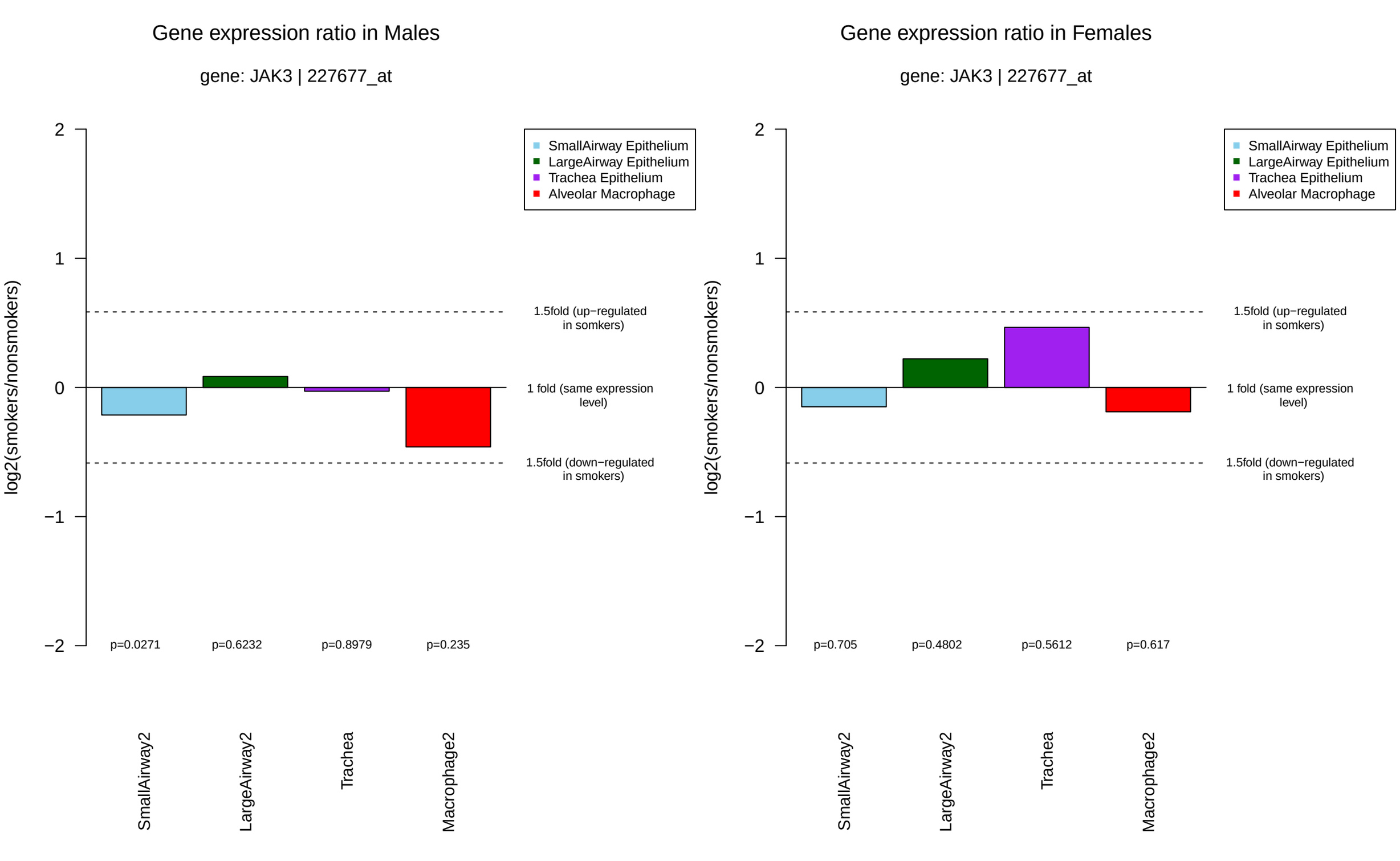
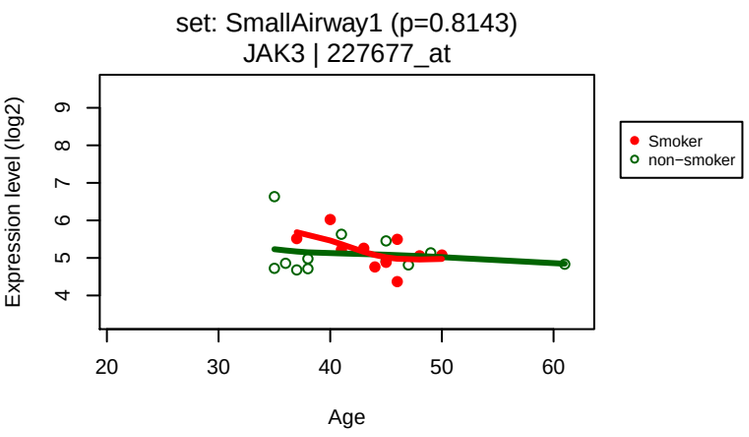
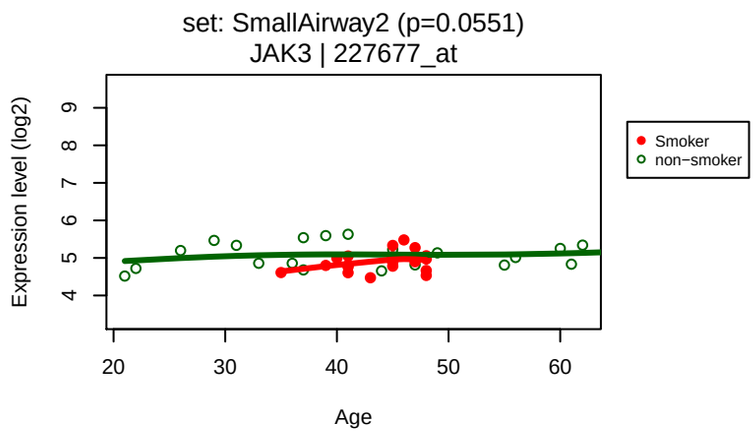
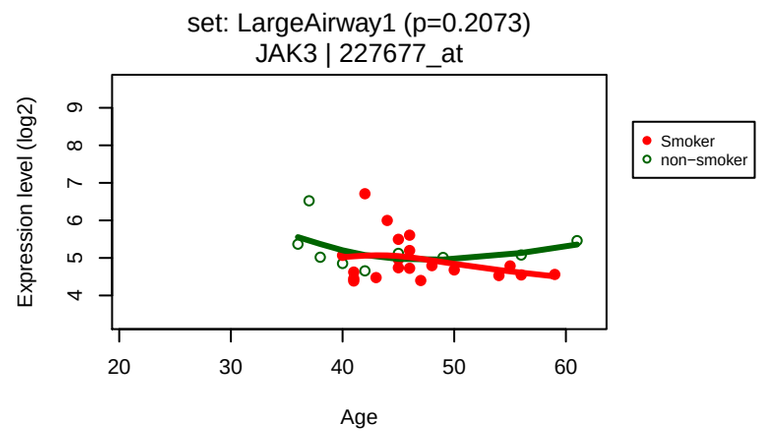

Gene expression of smokers and nonsmokers
gene: JAK3 | 227677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8143
p=0.0551
p=0.1273
p=0.2073
p=0.4607
p=0.6865
p=5e−04
p=0.2156
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

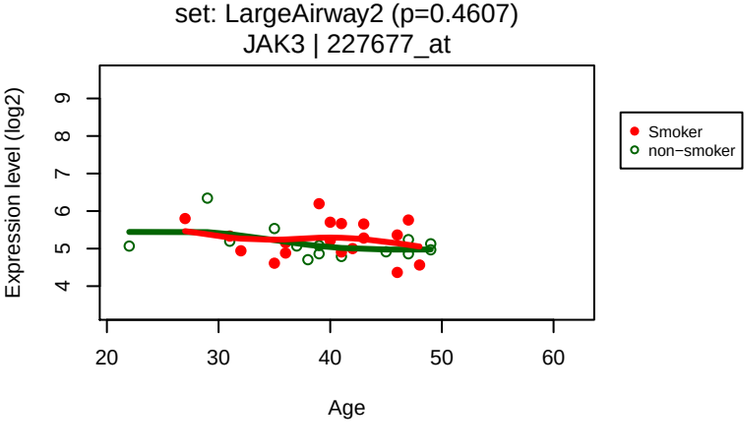

Gene expression ratio (smokers/nonsmokers)
gene: JAK3 | 227677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8143
p=0.0551
p=0.1273
p=0.2073
p=0.4607
p=0.6865
p=5e−04
p=0.2156
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

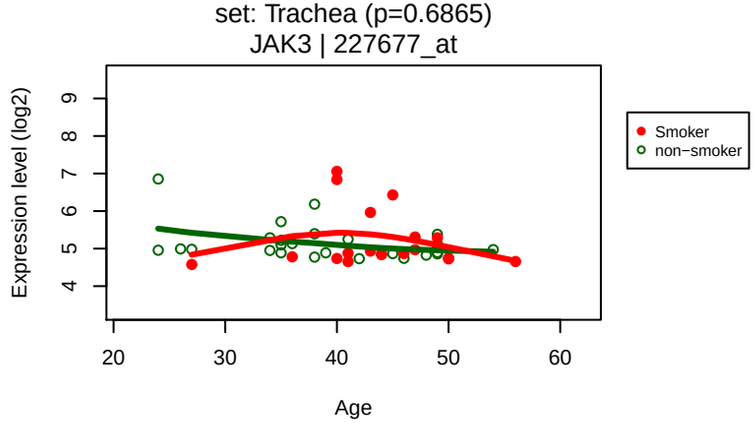

Gene expression ratio in Males
gene: JAK3 | 227677_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0271
p=0.6232
p=0.8979
p=0.235
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK3 | 227677_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.705
p=0.4802
p=0.5612
p=0.617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

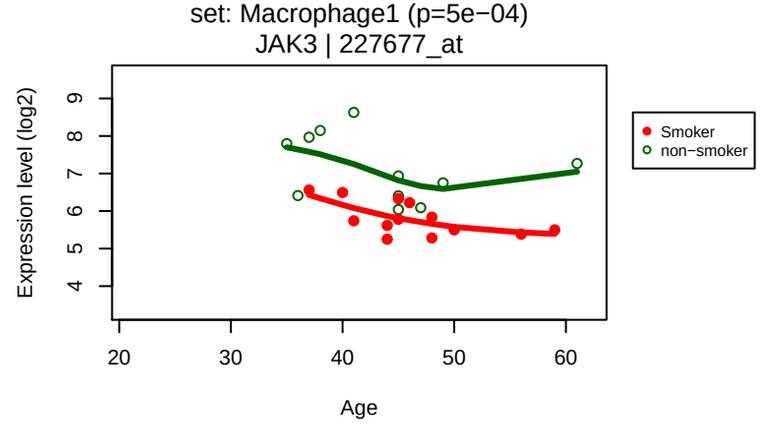

set: SmallAirway1 (p=0.8143)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker

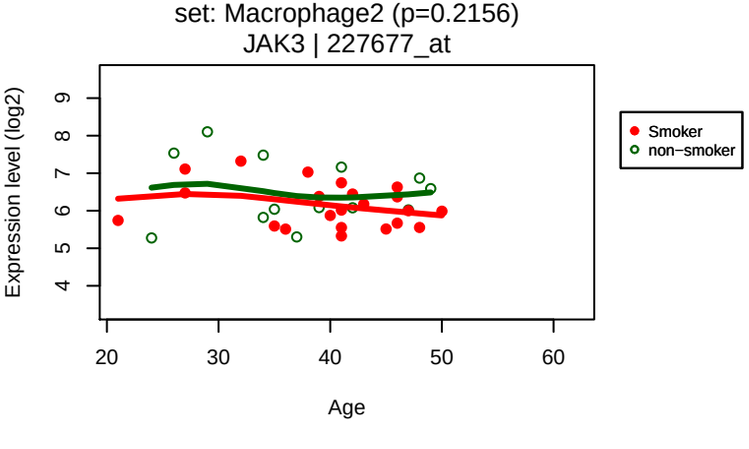

set: SmallAirway2 (p=0.0551)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker

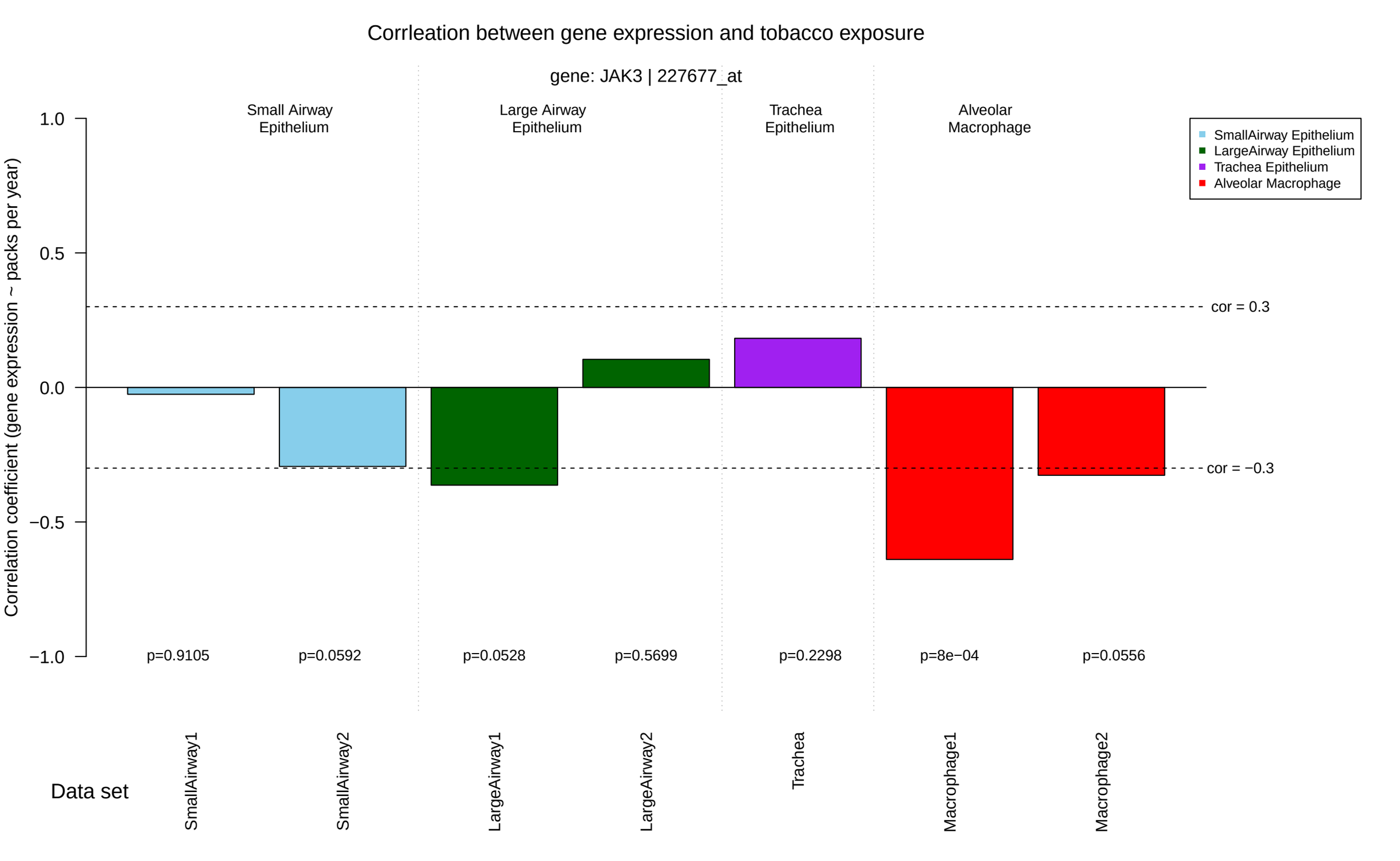

set: LargeAirway1 (p=0.2073)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4607)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6865)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=5e−04)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2156)
JAK3 | 227677_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK3 | 227677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9105
p=0.0592
p=0.0528
p=0.5699
p=0.2298
p=8e−04
p=0.0556
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2